

Q1. The reaction centre in PS II has an absorption maxima at:

NEET - 2023

1.	780 nm	2.	680 nm
3.	700 nm	4.	660 nm

Q2. How many ATP and NADPH₂ are required for the synthesis of one molecule of Glucose during Calvin cycle?

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1.	18 ATP and 16 NADPH ₂	2.	12 ATP and 12 NADPH ₂
3.	18 ATP and 12 NADPH ₂	4.	12 ATP and 16 NADPH ₂

Q3. Which of the following combinations is required for chemiosmosis? NEET - 2023

1.	Proton pump, electron gradient, NADP synthase.
2.	Membrane, proton pump, proton gradient, ATP synthase.
3.	Membrane, proton pump, proton gradient, NADP synthase
4.	Proton pump, electron gradient, ATP synthase

Q4. Match List - I with List - II

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List - I	List - II
(A) Chlorophyll a	(I) Yellow to yellow orange
(B) Chlorophyll b	(II) Yellow green
(C) Xanthophyll	(III) Blue green
(D) Carotenoids	(IV) Yellow

Choose the correct answer from the options given below:

Options:	(A)	(B)	(C)	(D)
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Q11. The products of light reaction in photosynthesis are:

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1. ATP , NADPH , and O₂
2. ATP , NADPH , O₂ and H₂O
3. ATP , NADPH And H₂O
4. ATP , NADPH and CO₂

Q12. The ratio of carbon dioxide fixation between C₄ plants and C₃ plants is :

1.	2:1	2.	2: 3
3.	1:1	4.	1:2

Q13. When one CO₂ molecule is fixed as one molecule of triose phosphate, which of the following photochemically made, high energy chemical intermediates are used in the reduction phase?

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1. 1 ATP + 1 NADPH
2. 1 ATP + 2 NADPH
3. 2 ATP + 1 NADPH
4. 2 ATP + 2 NADPH

Q14. Identify the correct statements regarding chemiosmotic hypothesis:

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a.	Splitting of the water molecule takes place on the inner side of the membrane.
b.	Protons accumulate within the lumen of the thylakoids.
c.	The primary acceptor of electrons transfers the electrons to an electron carrier.
d.	NADP reductase enzyme is located on the stroma side of the membrane.



4. PS-II to Cytb6f complex

Q18. The oxygenation activity of RuBisCo enzyme in photorespiration leads to the formation of:

NEET - 2020

1.	1 molecule of the 3-C compound
2.	1 molecule of the 6-C compound
3.	1 molecule of a 4-C compound and 1 molecule of the 2-C compound
4.	2 molecules of the 3-C compound

Q19. During non-cyclic photophosphorylation, when electrons are lost from the reaction center at PS II, what is the source which replaces these electrons? NEET - 2020

NEET - 2020

1.	Oxygen	2.	Water
3.	Carbon dioxide	4.	Light

Q20. Which of the following statements is incorrect?

NEET - 2020

1. RuBisCO is a bifunctional enzyme
2. In C₄ plants, the site of RuBisCO activity is mesophyll cell
3. The substrate molecule for RuBisCO activity is a 5-carbon compound
4. RuBisCO action requires ATP and NADPH

Q21. In Hatch and Slack pathway, the primary CO₂ acceptor is -

NEET - 2019

1. Oxaloacetic acid

2. Phosphoglyceric acid
3. Phosphoenol pyruvate
4. Rubisco

Q22. One scientist cultured *Cladophora* in a suspension of *Azotobacter* and illuminated the culture by splitting light through a prism. He observed that bacteria accumulated mainly in the region of:

NEET - 2019

1. Violet and green light
2. Indigo and green light
3. Orange and yellow light
4. Blue and red light

Q23. Which of the following is *not* a product of light reaction of photosynthesis?

NEET - 2018

1.	ATP	2.	NADH
3.	NADPH	4.	Oxygen

Q24. Oxygen is *not* produced during photosynthesis by:

NEET - 2018

1.	Green sulphur bacteria	2.	<i>Nostoc</i>
3.	<i>Cycas</i>	4.	<i>Chara</i>

Q25. With reference to factors affecting the rate of photosynthesis, which of the following statements is not correct?

NEET - 2017

- (1) Light saturation for CO₂-fixation occurs at 10% of full sunlight.
- (2) Increasing atmospheric CO₂ concentration upto 0.05% can enhance the CO₂-fixation rate
- (3) C₃ plants respond to higher temperature with enhanced photosynthesis, while C₄ plants have a much lower temperature optimum.
- (4) Tomato is a greenhouse crop, which can be grown in a CO₂-enriched atmosphere for higher yield.



Q26. Phosphoenol pyruvate (PEP) is the primary CO₂ acceptor in:

NEET - 2017

1. C₄ plants
2. C₂ plants
3. C₃ and C₄ plants
4. C₃ plants

Q27. Water vapour comes out from the plant leaf through the stomatal opening. Through the same stomatal opening carbon dioxide diffuses into the plant during photosynthesis. Reason out the above statements using the following options. **NEET - 2016**

1. Both process can happen together because the diffusion coefficient of water and CO₂ is different.
2. The above processes happen only during night time
3. One process occurs during the day time and the other night
4. Both processes cannot happen simultaneously

Q28. A plant in your garden avoids photorespiratory losses, has improved water use efficiency, shows high rates of photosynthesis at high temperatures and has improved efficiency of nitrogen utilization. In which of the following physiological groups would you assign this plant? **NEET - 2016**

- (1) C₄
- (2) CAM
- (3) Nitrogen-fixer
- (4) C₃

Q29. The process which makes a major difference between C₃ and C₄ plants is:

NEET - 2016

1. glycolysis	2. Calvin cycle
3. photorespiration	4. respiration

Q30. Water-soluble pigments found in plant cell vacuoles are:

NEET - 2016



1.	chlorophylls	2.	carotenoids
3.	anthocyanins	4.	xanthophylls

Q31. Emerson's enhancement effect and red drop have been instrumental in the discovery of?

NEET - 2016

1.	two photosystems operating simultaneously
2.	photophosphorylation and cyclic electron transport
3.	oxidative photophosphorylation
4.	photophosphorylation and non-cyclic electron transport

Q32. In chloroplast, the highest number of protons are found in?

NEET - 2016

1. lumen of thylakoids
2. intermembrane space
3. antennae complex
4. stroma

Q33. In photosynthesis, the light-independent reactions take place at? NEET - 2015

1. Thylakoid lumen
2. photosystem I
3. photosystem II
4. Stromal matrix

Q34. Anoxygenic photosynthesis is characteristic of:

AIPMT - 2014



- NEET - 2013
- expiration is: AIPMT - 2012

NEET - 2013

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AIPMT - 2012

- reticulum-dictyosomes
isome
e

AIPMT - 2011

Q38. In Kranz anatomy, the bundle sheath cells have:

AIPMT - 2011

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ANSWER KEY

1. (2)	11. (1)	21. (3)	31. (1)
2. (3)	12. (1)	22. (4)	32. (1)
3. (2)	13. (1)	23. (2)	33. (4)
4. (1)	14. (2)	24. (1)	34. (1)
5. (3)	15. (4)	25. (3)	35. (4)
6. (2)	16. (2)	26. (1)	36. (3)
7. (3)	17. (4)	27. (1)	37. (4)
8. (2)	18. (1)	28. (1)	38. (3)
9. (3)	19. (2)	29. (3)	39. (4)
10. (4)	20. (2)	30. (3)	40. (4)

